Now, answer the following questions.

**I. Here is your shopping list:**

**(i) 4 bottles of Protein Milk (200 ml size)** **(ii) 2 packets of Peanut candies(200 gm)**

**(iii) 1 packet of Chocolate biscuits** **(iv) 1 packet of Badam nuts (500 gm)**

(i) If you buy all the items in one shop, where will you get the best price?

(ii) If you buy the items from the two shops, how will you do it to spend the least amount of money?

**II. You have ₹1000/- to spend to buy the following shopping list:**

**(i) 6 bottles of Protein Milk (200 ml size)** **(ii) 3 packets of Peanut candies(200 gm)**

**(iii) 3 packets of Chocolate biscuits** **(iv) 1 packet of Badam nuts (250 gm).**

(i) How can you do this so that you don't go over your budget amount ₹1000?

(ii) Which shop offers you the best value for money on each item?

(iii) Is the "**BUY ONE GET ONE**" deal at Super Provisions the same as "**50% off**" deal?

## 7.9 Packing




When we are packing something in a box or a suitcase or a cupboard, first we have to decide how we are going to pack. How many items can be fitted into that fixed space? A good example of this is, before you go to school, you try to pack everything you need (like your books, notebooks, geometry box, sports equipment, food and water bottle) into your school bag. At that time, you are very clear that your books should not be damaged and you are able to carry everything yourself. Think! The same rules apply to posting a package to a friend or family member or others.

Apart from these, the packaging methods are used in many cases such as cutting of sheets, glass, paper, wood, cloth or other materials and room allotment, seating arrangement in the particular space provided, parking vehicles with proper lanes and saving data in hard disk, CD, pen drive and so on.

Using some of the packing methods from the following situations and examples, let us try to understand how best to fit the items into the space in the containers or in rooms or in boxes etc.,

### 7.9.1 Packing Approaches - Fractional method

Here, when we fill items in bags or in containers, we determine the weight, value and number of each item with the condition that the total weight of the container is less than or equal to a given limit and the total value is as large as possible. Fractional method uses the technique of buying things fractionally and admits buying of more items within a given budget. Let us learn more about this approach from the following situation.






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